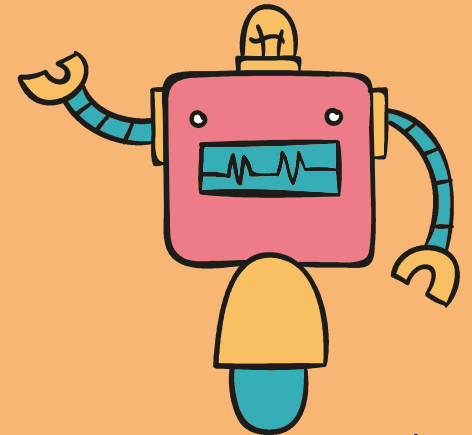
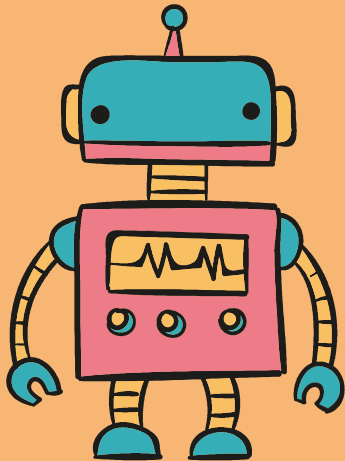
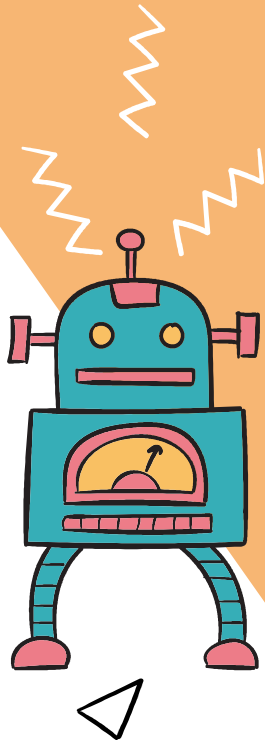


HUMAN FOLLOWING ROBOT





VISHWAKARMA INSTITUTE OF TECHNOLOGY,
PUNE.



01

Department

Department of
Engineering Sciences
and Humanities.

02

PROJECT GUIDE

Prof. Swati Joshi

03

MEMBERS

- Nikhil Yeware 53
- Yukti Ahirkar 54
- Yuteeka Kamath 55
- Yuvraj Susatkar 56
- Komal Zabak 57
- Mahek Zade 58

04

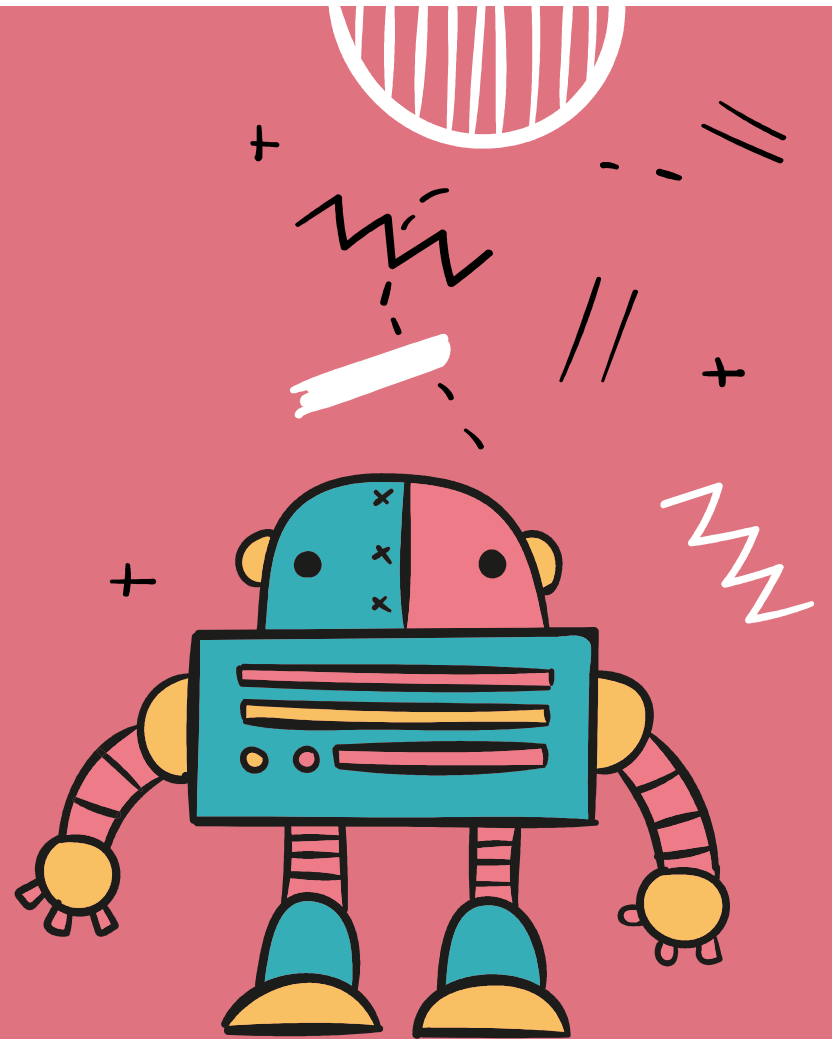
BATCH/GROUP

B3/T11

Introduction

Robotic technology has increased appreciably in past couple of years. Such innovations were only a dream for some people a couple of years back. But in this rapid moving world, now there is a need of robot such as “A Human Following Robot” that can interact and co-exist with them.

Tasks performed by humans have its own limitations in many ways. In order to perceive beyond the human limitation in vision, speed, consistency, flexibility, quality e.tc we should make use of robots.

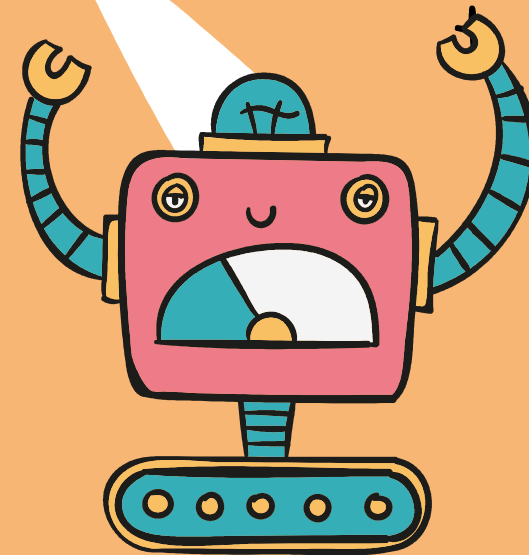


Problem Statement

Creating an efficient and portable mobile robot system which can operate on human commands.

The selected region gets jumped from the user to an obstacle when the user is close to this obstacle.

When a user is moving fast and away from the robot, the depth value of the user is jumping from one frame to the other.

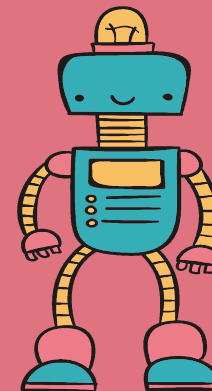




Motivation

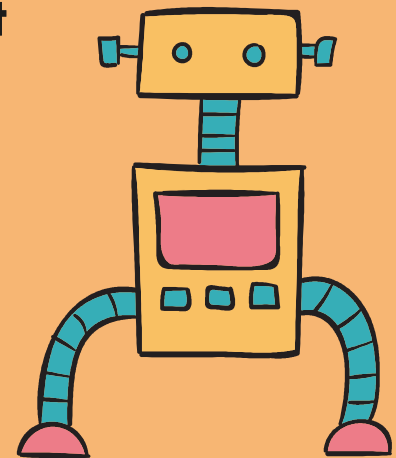
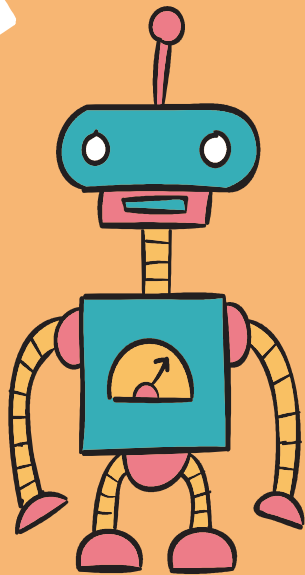
This project is built as a blueprint for a fully functional robot system which can respond to human gestures and commands.

Teamwork with robots often involves both physical and mental activities. This implies that motivation might be particularly important to the success of human robot interaction.



LIMITATIONS

The last problem encountered is the fact that, as the robot is not very tall, most of the time the robot was seeing the ground. If the user is far from the robot, the ground is then merged with the region corresponding to the user.



Literature Reviews

	Title	Summary
Review 1	HUMAN FOLLOWING ROBOT USING ARDUINO UNO	In this paper the robot has Infrared sensors used to move it in both the direction and ultrasonic sensor for both the forward and reverse direction. We used the Arduino Uno microcontroller as the brain of this project. This robot is driven with four Dc motors and it is controlled by a motor driver shield.
Review 2	STUDY OF ULTRASONIC SENSORS IN HUMAN FOLLOWING ROBOT	This paper discusses merits of ultrasonic sensors in robot, which is used for human presence detection. In this robot 3 ultrasonic sensors of 40Khz frequency are used for detection and the robot technology has been tested in many different angles of the sensors.
Review 3	STUDY OF VISION AND OBJECT TRACKING IN HUMAN FOLLOWING ROBOT USING ULTRASONIC AND INFRARED SENSORS	This paper discusses about the success and problems encountered during vision and object tracking in a human following robot. It specifically tells about the algorithms, sensors, circuit arrangement, sensor calibration and microcontroller operations.

Literature Reviews

Title

Summary

Review 4

HUMAN FOLLOWING ROBOT

In the paper Arduino IDE application are used it's act as a processor and gives output to motor driver. The RFID tag is used to tag the owner of the robot and follow the person without getting mis leaded by any other person It can be provided with sensors like Ultra sonic and IR sensor for obstacle detection and following human.

Review 5

DESIGN AND DEVELOPMENT OF HUMAN FOLLOWING ROBOT

In this paper, we presented a method of a human following robot based on tag identification and detection by using a camera. Intelligent tracking of specified target is carried out by the use of different sensors and modules ultrasonic sensor, magnetometer, infrared sensors and camera.

An intelligent decision is being made by the robot control unit based on the information obtained from the above sensors and modules, hence finding and tracking the particular object by avoiding the obstacles and without collision with the target.

Requirements



Software

- Arduino Uno IDE

Platform

- Mac OS
- Window7,10,11



Hardware

- Arduino Uno
- L293D Motor driver
- Infrared Sensors
- Ultrasonic Sensor
- Servo Motor
- Four DC Geared Motors
- Four Wheels
- Robot Chassis
- Jumper Wires
- 18650 Batteries
- Switch
- Other Tools

Requirements



Arduino Uno

It is the brain of our project. It can give all the command to their sub ordinate components which should be operated by the human behavior. And it also give feedback to the other components and human. So that it can be used as a medium of communication between human and robots & vice versa.



DC Motors

DC Motor is a device that converts any form of energy into mechanical energy or imparts motion. In constructing a robot, motor usually plays an important role by giving movement to the robot. Here 4 DC motor are used to drive the robot.



Motor Shield

Motor Shield The Motor Shield is a driver module for motors that allows you to use Arduino to control the working speed and direction of the motor. The Motor Shield can either be powered by Arduino directly or by an external 6V~15V power supply via the terminal input.



Ultrasonic sensor

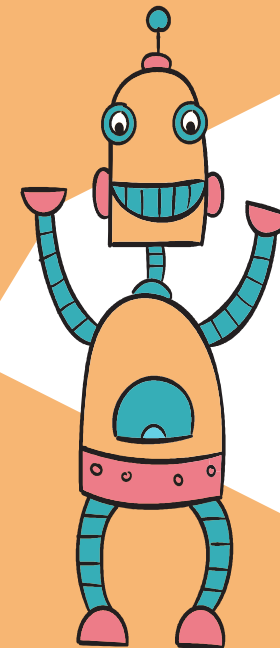
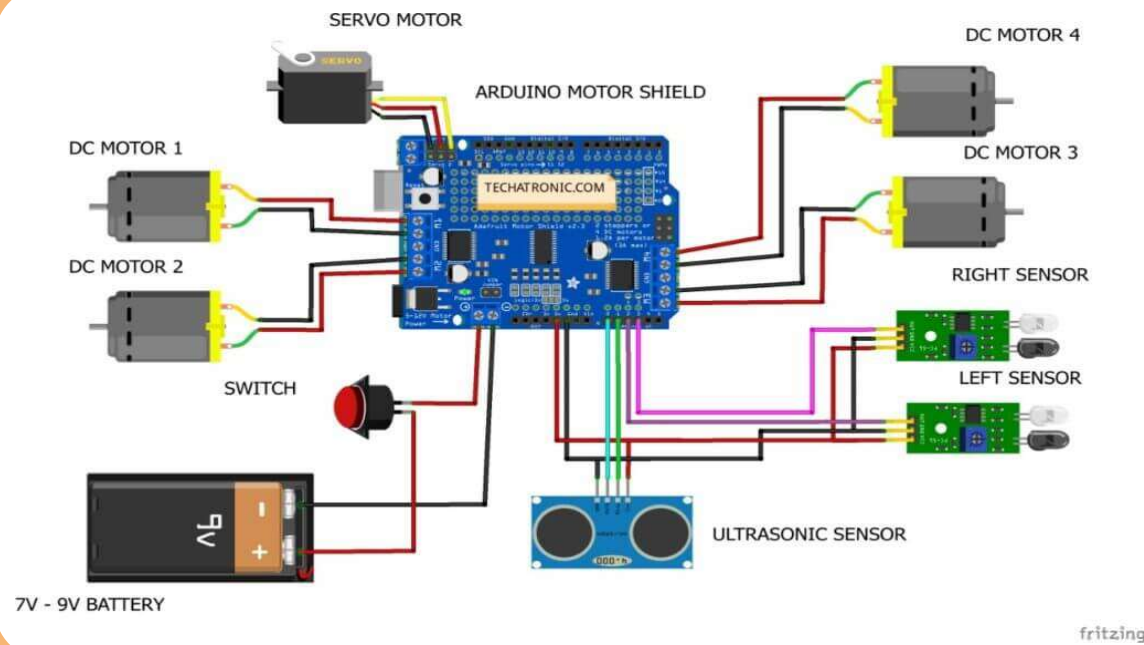
An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. The working principle of this module is simple, it sends an ultrasonic pulse out at 40kHz which travels through the air and if there is an obstacle or object, it will bounce back to the sensor. By calculating the travel time and the speed of sound, the distance can be calculated.

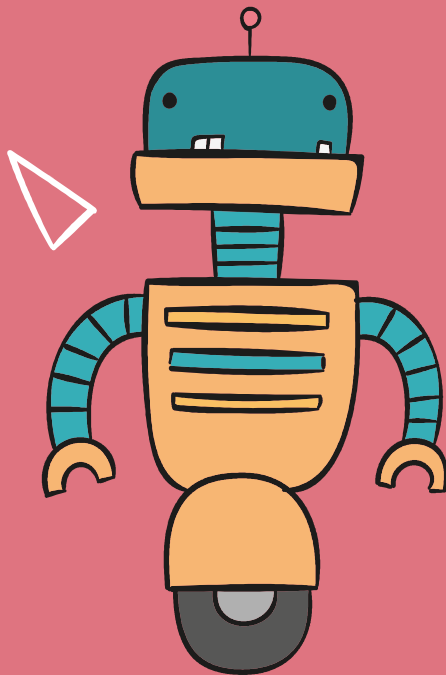


IR Sensor

IR sensor is an electronic device, that emits the light in order to sense some object of the surroundings. An IR sensor can measure the heat of an object as well as detects the motion. Usually, in the infrared spectrum, all the objects radiate some form of thermal radiation. These types of radiations are invisible to our eyes, but infrared sensor can detect these radiations. The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode.

Circuit Diagram



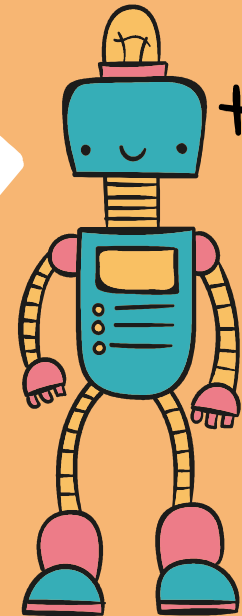


Future Scope

There are many interesting applications of this research in different fields whether military or medical. A wireless communication functionality can be added in the robot to make it more versatile and control it from a large distance. This capability of a robot could also be used for military purposes. By mounting a real time video recorder on top of the camera, we can monitor the surroundings by just sitting in our rooms. We can also add some modifications in the algorithm and the structure as well to fit it for any other purpose. Similarly it can assist the public in shopping malls. So there it can act as a luggage carrier, hence no need to carry up the weights or to pull that. Similarly, ample amount of modifications could be done to this prototype for far and wide applications.

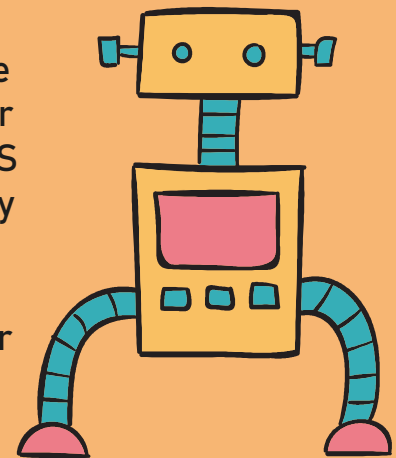
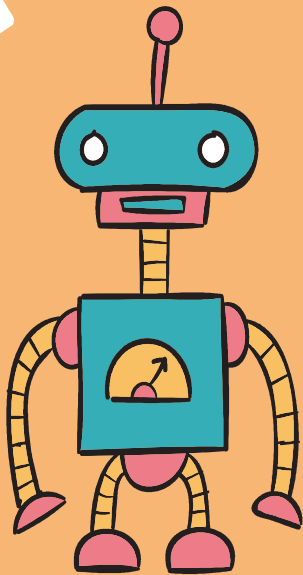
Conclusion

The human following robot is an automobile system that has ability to recognize obstacle, move and change the robot's position toward the subject in the best way to remain on its track. This project uses arduino, motors different types of sensors to achieve its goal. This project challenged the group to cooperate, communicate, and expand understanding of electronics, mechanical systems, and their integration with programming.



Acknowledgement

No volume of words is enough to express our gratitude towards our guide mam Mrs. Swati Joshi mam ,who has been very concerned and guided for all the ideas and solve quires in the preparation of this project. They have been very cooperative and sincere us through to see our hard work become fruitful. We are very thankful to Mr.SS Sawant project coordinator for giving us the opportunity to carry out this project . We also want to express our thanks to our parents for giving us the time and money for this project. And finally, we would like to express our gratitude towards our team members for giving their 100% effort and their hard work.



**THANK
YOU.**

